



“THE IMPACT OF **AL-JAZARI'S INVENTIONS** IS STILL FELT IN **MODERN CONTEMPORARY MECHANICAL ENGINEERING.**”

Donald Hill, from *Studies in Medieval Islamic Technology*, 1998

ITALIAN MATHEMATICIAN

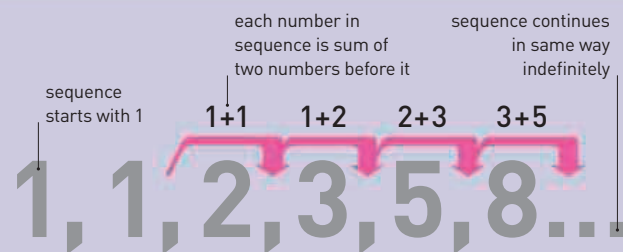
Leonardo Pisano (Fibonacci) published the *Liber Abaci* (*Book of Calculations*) in 1202, the first major western European work popularizing the use of **Hindu-Arabic numerals** and **place notation** (see 861–99). The book also presented **rules for algebra**, which he probably derived from al-Khwarizmi (see 821–60), as well as solutions for finding square and cube roots. Fibonacci described techniques

Ingenious devices

This illustration of one of al-Jazari's mechanical devices shows an automaton that pours water from a pot, then returns to a chamber where it is scooped back up again.

that were useful for the Pisan merchants of his day, including a method for multiplication using a grid, advice on the barter of goods, and the use of alloys to make coins. The Fibonacci sequence (below) is derived from a problem that concerned the growth of a rabbit population.

In 1206, Arab engineer **Ibn Isma'il al-Jazari** published the *Book of Knowledge of Ingenious and Mechanical Devices*, detailing 50 machines, including the **first descriptions of crankshafts and camshafts**. The most spectacular of these was a 2 m- (6.5 ft-) high water clock in the form of an elephant with a phoenix that marked half-hours.



FIBONACCI SEQUENCE

The Fibonacci sequence is a series of numbers in which each successive number is the sum of the two numbers preceding it. Any number in the series is known as a Fibonacci number. These occur surprisingly often in nature, with the number of petals of many flowers being Fibonacci numbers (daisies have 13, 21, or 34), while the arrangement of leaves on a plant stem is also determined according to a ratio connected to this sequence.

1202 Fibonacci publishes the *Liber Abaci* expounding use of Hindu-Arabic numerals

1206 Al-Jazari's *Book of Knowledge of Ingenious and Mechanical Devices* describes crankshafts and camshafts

1210 A church synod bans the study of the works of Aristotle at Paris University

1214 Italian physician Hugh of Lucca uses wine as an antiseptic and observes role of pus in infection

1217 Scottish scholar Michael Scot translates al-Bitruji's *Planetary Theory* into Latin